

Looking for edges

If you have used any sophisticated image editing program (such as Photoshop, GIMP, Paint.NET etc), you would know of an effect which highlights edges in the pictures. A similar approach serves really well in detecting



The process of detecting line of sight

objects. Unlike gradients which vary under lighting conditions, edges do not. In most normal lighting conditions, location and size of edges do not change.

Comparing the edges in the image received via the camera with entries in the database of images with similar

edge patterns can reveal the nature of object. For example, the edge patterns of a table are going to be different than that of a chair. Change in the position of viewing angle however requires that the edge templates of known objects should also be available from various angles.

Search in large databases of object models

This is more like an exhaustive search which would be done on a large set of object models. In this method, the image is submitted for a search in a database which contains the models for various objects from multiple angles. This method is synonymous to the exhaustive search method in other realms of data management and processing.

Feature interpretation search

If you aren't clear with data structure concepts yet, this might not make much sense. In this method, first of all, features are searched for in the image e.g. search for a 'window', a 'door' and a 'chimney outlet' in a picture of a 'house' should yield hits. Thereafter, comparisons are run for the identical placements of such features in the database of known objects. In our example, the object might just get recognized as a house. The search is done on the nodes of a feature tree. The root node of this tree is kept empty (indicating that no feature was matched). All child nodes in the tree contain all features found in the parent plus one more. As more and more shapes are found, the traversal goes down the tree to find the correct feature set identified in the image.